

Simple harmonic motion with damping

Basic skills

Q1. a) $\ddot{x} = -\omega^2 x$

If spring under tension $\omega = \sqrt{\frac{k}{m}}$

b) solutions

$m = \pm \omega i$

$$x(t) = A \cos \omega t + B \sin \omega t$$

Q2. $k = 30, m = 3$ so $\omega = \sqrt{\frac{30}{3}} = \sqrt{10}$

and $x(0) = 0.3, \dot{x}(0) = 0$

In m

$$x(t) = A \cos \sqrt{10}t + B \sin \sqrt{10}t$$

$$x(0) = 0.3 \Rightarrow A = 0.3$$

$$\dot{x}(t) = -\sqrt{10} A \sin \sqrt{10}t + \sqrt{10} B \cos \sqrt{10}t$$

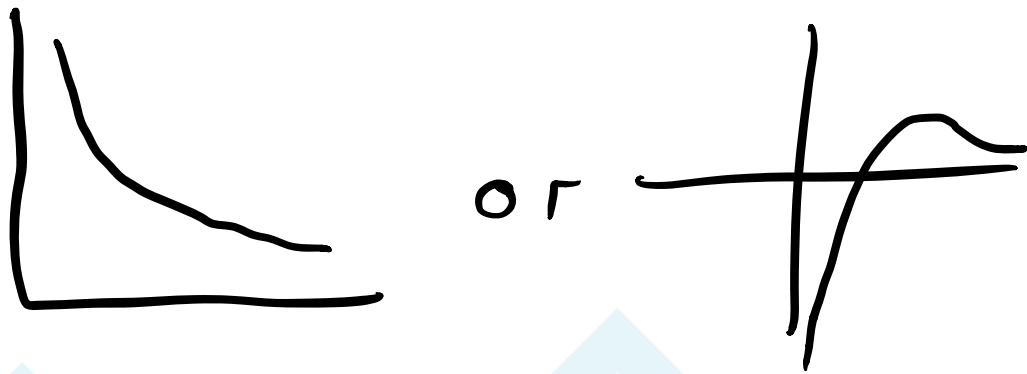
$$\dot{x}(0) = 0 \Rightarrow \sqrt{10} B = 0 \Rightarrow B = 0$$

so $x(t) = 0.3 \cos \sqrt{10}t$

Q3. Light damping - Decays with oscillations slowly



Heavy damping - decays without oscillations



critical - similar to heavy



Competency skills

Q4. $d = b^2 - 4ac$ aux sol: $m = \frac{-b \pm d}{2a}$

$d > 0$ $Ae^{m_1 x} + Be^{m_2 x}$

for zero at infinity $m_1, m_2 < 0$ so $b > 0$

$d = 0$ $Ae^{mx} + Bxe^{mx}$

for zero at infinity $m < 0$ so $b > 0$

$d < 0$ $e^{\text{Re}(m)x} (A \cos(\text{Im}(m)x) + B \sin(\text{Im}(m)x))$

for zero at infinity $\text{Re}(m) < 0$ so $b > 0$

Q5. $\ddot{x} + k\dot{x} + 4x = 0$, $x(0) = 0$
 $\dot{x}(0) = 1$

a) $k = 5$

aux: $m^2 + 5m + 4 = 0$ $(m+1)(m+4) = 0$

$m = -1, -4$ so $x(t) = Ae^{-t} + Be^{-4t}$

$x(0) = 0 \Rightarrow A + B = 0$

$\dot{x}(0) = 1 \Rightarrow -A - 4B = 1 \Rightarrow -3B = 1$
 $B = -\frac{1}{3}$
 $A = \frac{1}{3}$

so $x(t) = \frac{1}{3}(e^{-t} - e^{-4t})$

↑ Heavy damping

b) $k = 2$

aux: $m^2 + 2m + 4 = 0$ $m = \frac{-2 \pm \sqrt{12}}{2} = -1 \pm \sqrt{3}i$

$x(t) = e^{-t}(A \cos \sqrt{3}t + B \sin \sqrt{3}t)$

$x(0) = 0 \Rightarrow A = 0$

$\dot{x}(0) = 1 \Rightarrow \sqrt{3}B = 1$ $B = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$

so $x(t) = \frac{\sqrt{3}}{3}e^{-t} \sin \sqrt{3}t$

↑ Light damping

c) $k=4$

aux: $m^2 + 4m + 4 = 0 \quad (m+2)^2 = 0 \quad m = -2$

so $x(t) = Ae^{-2t} + Bte^{-2t}$

$x(0) = 0 \Rightarrow A = 0$

$\dot{x}(0) = 1$ as $\dot{x}(t) = Be^{-2t} - 2Bte^{-2t}$

$\Rightarrow B = 1$ so $x(t) = te^{-2t}$

critical damping

Q6. $\ddot{x} - 5\dot{x} + 4x = 0$ is not damped
as $k = -5 < 0$ as solution is

$x(t) = Ae^t + Be^{4t}$ exponential growth.

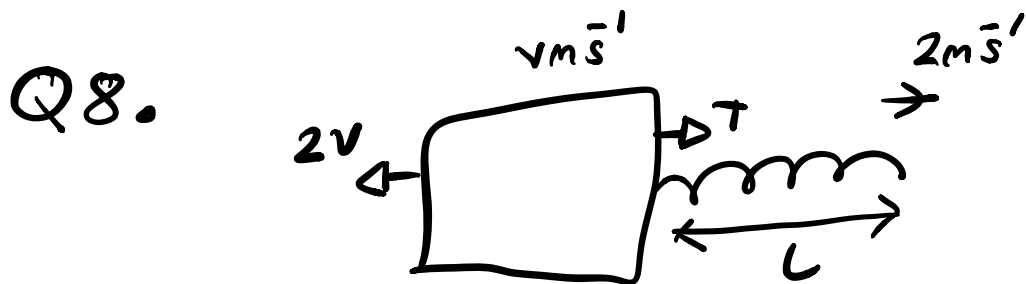
Q7. $\omega = \sqrt{\frac{50}{2}} = 5 \quad \ddot{x} + k\dot{x} + 5^2x = 0$

For critical damping all need is

$k > 0$ and $d = k^2 - 4 \times 5^2 = 0$

$\Rightarrow k^2 = 100 \quad k = \sqrt{100} = 10$

only positive



a) $T = k(L - 0.4)$ by Hooke's law

so $F = T - 2v = k(L - 0.4) - 2v$

b) As length changes depending on change in positions of spring ends so

$$\frac{dL}{dt} = \underset{\substack{\uparrow \\ \text{Front}}}{2} - \underset{\substack{\uparrow \\ \text{Back}}}{v}$$

c) $40a = k(L - 0.4) - 2v$

As acceleration of block is opposite of the change in rate of length

so $a = -\frac{d^2L}{dt^2} \Rightarrow -40 \frac{d^2L}{dt^2} = kL - 0.4k - 2v$

$$40 \frac{d^2L}{dt^2} = 2v - kL + 0.4k$$

$$\text{as } v = 2 - \frac{dL}{dt} \Rightarrow 40 \frac{d^2L}{dt^2} = 4 - 2 \frac{dL}{dt} - kL + 0.4k$$

$$\Rightarrow \frac{d^2L}{dt^2} + 0.05 \frac{dL}{dt} + \frac{k}{40} L = 0.1 + 0.01k$$

$$d) \text{ aux } m^2 + 0.05m + 0.0004 = 0$$

$$(m + 0.04)(m + 0.01) = 0$$

$$m = -0.01, -0.04$$

$$L = A e^{-0.01t} + B e^{-0.04t}$$

$$\text{For Particular. } \ddot{L} + 0.05\dot{L} + 0.0004L = 0.10016$$

$$\text{So } L = \text{const.} \quad \begin{aligned} 0.0004a &= 0.10016 \\ a &= 250.4 \end{aligned}$$

$$\text{So } L = 250.4$$

$$\text{So } L = A e^{-0.01t} + B e^{-0.04t} + 250.4$$